CS215: DATA ANALYSIS & INTERPRETATION

Checkpoint 1 Progress Report & Meta Information

SECTION 1: META INFORMATION

Project Title:
Statistical Analysis of Home-Ground Advantage in the IPL

Group Number:
46

Mentor Name:
Yugesh

GitHub Repository Link:
https://github.com/Chetandoescode7781/Statistical-Analysis-of-Home-Ground-Advantage-in-IPL

Member Short Name	Project Role
Chetan	Data Collection & Cleaning Lead
Vinay	Probability Modeling & Win Rates
Ankit	Batting & Bowling Metrics Specialist
Satyam	Inference, Hypothesis Testing & BI

SECTION 2: PROGRESS SO FAR

2.1 Descriptive Statistics of the Dataset(s)

We have successfully integrated two comprehensive datasets spanning all 19 seasons of the Indian Premier League (2008–2026). The datasets are publicly accessible and licensed for open research usage.

1,100+	260,000+	19	35+
TOTAL MATCHES	BALL-BY-BALL RECORDS	SEASONS (2008-2026)	UNIQUE VENUES

Dataset Identifier & Source	Rows	Cols	Key Attributes	Access / Permissions
Ritesh Ojha IPL Dataset (GitHub Repository)	~1,095	20	match_id, season, venue, team1, team2, toss_winner, winner, win_by_runs	Public Open License
Kaggle IPL Ball-by-Ball 2008–2026 (All 19 Seasons Dataset)	~260,000	22	match_id, inning, over, ball, batter, bowler, runs_off_bat, extras, venue	Public Domain / Kaggle API

2.2 Data Cleaning Steps Completed

- **Franchise Name Standardization:** Harmonized team name changes across 19 seasons (e.g., mapping *Royal Challengers Bangalore → Royal Challengers Bengaluru*, *Delhi Daredevils → Delhi Capitals*, and *Deccan Chargers → Sunrisers Hyderabad*).
- **Home Ground Mapping:** Mapped home grounds for each core franchise (e.g., Eden Gardens → KKR, M. Chinnaswamy → RCB, Wankhede → MI, Chepauk → CSK) and classified neutral/shifted venues (e.g., COVID-19 UAE phases).
- **Missing Data & Tie Handling:** Imputed missing venue metadata and standardized outcome flags for tied matches resolved via Super Overs.

2.3 Division and Revision of Responsibilities

The initial work plan was refined to accommodate team strengths and complex data pre-processing requirements for ball-by-ball analysis:

Member	Original Proposed Scope	Revised Responsibility & Workflow	Status
Chetan	Data collection & basic statistics	Standardize 19-season team/venue data, build automated Kaggle API ingestion pipeline, and compute home vs. away sample metrics.	Completed
Vinay	Match outcomes & Bernoulli modeling	Estimate home/away win probabilities, fit Bernoulli models for match outcomes, and conduct binomial distribution comparisons across venues.	In Progress
Ankit	Batting and bowling performance	Extract over-by-over run rates, bowling economy, and powerplay/death-over splits across home and away grounds using pandas.	In Progress
Satyam	Statistical inference & visualization	Execute two-sample t-tests/chi-square tests, construct 95% confidence intervals, and synthesize results into an interactive Power BI dashboard.	In Progress

2.4 Fine-Grained Problem Statement Refinement

While our primary inquiry evaluates whether playing on home grounds yields a statistically significant win probability increase, we have refined our analysis into three sub-hypotheses:

- **H1 (Overall Win Rates):** Home teams exhibit a win probability $P(\text{Win} \mid \text{Home}) > 0.50$ across all 19 seasons using a one-sided Z-test for proportions.
- **H2 (Scoring & Pitch Advantage):** Mean first-innings scores on home grounds are significantly higher than overall venue averages due to familiar pitch conditions ($p < 0.05$).
- **H3 (Franchise Variance):** Home advantage is non-uniform across teams—certain teams (e.g., CSK at Chepauk) exhibit a disproportionately higher home win rate compared to others.

2.5 Updated Tracking Spreadsheet & Meta-Information Row

Our tracking sheet has been updated with detailed task breakdowns, milestone dates, and mentor access permissions.

> **Tracking Sheet Link:** Google Sheets - CS215 Team 46 Tracking Document
> *Note: Verified full **EDIT permissions** granted for Mentor and CS215 Teaching Assistants. The top row contains project meta-information (Title, Group #46, Repository, and Member Roll Numbers).*